

**MAHATMA EDUCATION SOCIETY'S
PILLAI COLLEGE OF ARTS, COMMERCE & SCIENCE**
(Autonomous)
NEW PANVEL

PROJECT REPORT ON

**“SOCIAL MEDIA MARKETING
AGENCY SYSTEM”**

**IN PARTIAL FULFILLMENT OF
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PROJECT GUIDE

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INTRODUCTION

Overview

A social media marketing agency is a business that helps other businesses promote their products and services on platforms such as Instagram, Facebook, YouTube and LinkedIn. It designs posts, edits reels, runs paid advertising campaigns and manages the social media accounts of its clients. Every client chooses a platform, a service and a monthly budget, so the agency must keep an accurate record of all its clients.

The Social Media Marketing Agency System is a Python-based console application that keeps this client record. After a staff member logs in, the system allows them to add new clients, display all clients with the total monthly budget, search any client by name or phone number, update a client's service package and delete a client. All records are stored in a file, so they remain available every time the program is started again.

Linked List in Python

The client list is created using the Linked List data structure. A linked list provides a logical connection between data elements that are stored in memory at different locations physically. Each data element is stored in a node along with a reference link to the next node. The first node is called the head and the link of the last node is None. The entire list is accessed sequentially by starting from the head and moving to the next node until the end is reached.



Figure 1: Client records stored as nodes of a singly linked list

SWOT ANALYSIS

SWOT analysis studies the Strengths and Weaknesses inside a business and the Opportunities and Threats outside it. The SWOT analysis of a social media marketing agency is given below.

STRENGTHS (S) • Low start-up cost – needs only computers, internet and a skilled team. • Can reach a large and exactly targeted audience by age, city and interests. • Results are measurable – reach, likes, clicks and leads can be shown to clients. • Many services in one place: post design, reels, ads and account management. • Monthly packages give the agency a regular income.	WEAKNESSES (W) • Depends on platforms it does not control; an algorithm change can cut reach overnight. • Needs fresh creative content every day, which is hard to keep up. • Clients expect quick results, but building a brand takes time. • A small agency depends on a few clients; losing one affects income. • Client records kept on paper or in chats lead to mistakes and missed follow-ups.
OPPORTUNITIES (O) • Small and local businesses (shops, cafés, gyms) are moving online. • Short videos (Reels, Shorts) and influencer marketing are growing fast. • Regional-language content for smaller cities and towns. • AI tools for designs, captions and scheduling save time. • Online stores need paid ads to get sales.	THREATS (T) • Heavy competition from other agencies and low-cost freelancers. • Businesses can use free tools (Canva, Meta Business Suite) and do it themselves. • Sudden changes in platform rules or advertising prices. • Data privacy laws limit how customer data can be used for targeting. • In a slowdown, marketing budgets are the first to be cut.

LEARNING FROM THE CASE

From the Social Media Marketing Agency Case

1. An agency's biggest asset is its client list. Every client's platform, service and budget must be recorded correctly and be easy to find.
2. Depending on one platform or one big client is risky; the agency should spread its work across platforms and clients.
3. Showing clear, measurable results (reach, leads, sales) keeps clients coming back every month.
4. The market changes quickly – reels, influencers, AI tools and new rules – so the agency must keep learning and adapting.
5. Knowing the total monthly budget of all clients helps the agency plan its staff and income.

From Building the System

6. A linked list grows as clients are added. There is no fixed size to decide in advance, unlike an array.
7. Display, search, update and delete all work by traversing the list from the head, one node at a time. Searching this way is called linear search.
8. Deleting a client only changes one link (`previous.next = current.next`). No elements have to be shifted as in an array.
9. File handling makes the data permanent: the list is built from `clients.txt` when the program starts and saved back to the file after every change.
10. Checking the login from `login.txt` is better than writing usernames and passwords inside the code.
11. Checking the input (10-digit phone, numeric budget, valid menu choice) stops wrong data from being stored.

OBJECTIVE OF THE PROJECT

- To develop a simple system for keeping the records of all clients of a social media marketing agency.
- To enable easy adding, searching, updating and deleting of client records.
- To show all clients along with the total monthly budget of the agency.
- To implement dynamic data management using the Linked List data structure.
- To store the records permanently using file handling and to check the login from a file.
- To offer a menu-driven interface for easy user interaction.

FEATURES OF THE PROJECT

1. **Secure Login:** The username and password are checked against login.txt; only 3 attempts are allowed.
2. **Add Client:** A new client is stored as a node at the end of the linked list. The platform and service are chosen from a menu, and a duplicate client name is not allowed.
3. **Display Clients:** All clients are shown in a table with the total number of clients and the total monthly budget.
4. **Search Client:** Any client can be found by name (in any letter case) or by phone number.
5. **Update Package:** The service and monthly budget of a client can be changed.
6. **Delete Client:** A client is removed by unlinking its node from the list.
7. **Permanent Storage:** Clients are loaded from clients.txt at the start and saved after every change.
8. **Input Validation:** The system re-asks for an invalid phone number, platform, service or budget and rejects invalid menu choices.

TECHNOLOGIES USED

The project is entirely developed using Python, with key concepts such as:

- **Linked List** (custom Client node class) – To store all clients dynamically.
- **Python List** – To store the fixed choices of platforms and services.
- **File Handling** (open, read, write) – To store client records and login details permanently.
- **Menu-driven interface** – For user interaction via input().
- **Conditional Statements & Loops** – To control program flow and validations.
- **Exception Handling (Basic)** – Handles a missing clients.txt or login.txt file.
- **String methods** (split, strip, lower, isdigit) – To read file lines and check the input.

SCOPE OF THE PROJECT

This project can be extended in the digital marketing industry to include:

- Graphical User Interface (GUI) or a web dashboard for easier access.
- Database integration to store a large number of client records.
- Sorting clients by name or budget using sorting algorithms on the linked list.
- Monthly invoices and payment tracking for every client.
- Encrypted passwords and different roles for staff members.
- Campaign analytics (reach, likes, leads) for every client through platform APIs.

CODE

```
# SOCIAL MEDIA MARKETING AGENCY SYSTEM
# Data Structure used : Singly Linked List
# Files used          : clients.txt -> client records
#                      (name,phone,platform,service,budget)
#                      login.txt   -> staff login details (username,password)

CLIENT_FILE = "clients.txt"
LOGIN_FILE = "login.txt"

PLATFORMS = ["Instagram", "Facebook", "YouTube", "LinkedIn"]
SERVICES = ["Post Design", "Reels Editing", "Ad Campaign", "Account Management"]

class Client:
    # one node of the linked list = one client of the agency
    def __init__(self, name, phone, platform, service, budget):
        self.name = name
        self.phone = phone
        self.platform = platform
        self.service = service
        self.budget = budget
        self.next = None

class AgencyList:
    def __init__(self):
        self.head = None

    # insert a new client node at the end of the list
    def add_client(self, name, phone, platform, service, budget):
        new_node = Client(name, phone, platform, service, budget)
        if self.head is None:
            self.head = new_node
            return
        current = self.head
        while current.next is not None:
            current = current.next
        current.next = new_node

    # visit every node from head to the end and print it
    def display_clients(self):
        if self.head is None:
            print("--- No clients available ---")
            return
        print(f"{'NO':<4}{'NAME':<19}{'PHONE':<12}"
              f"{'PLATFORM':<11}{'SERVICE':<20}{'BUDGET':>8}")
        print("-" * 74)
        count = 0
        total = 0
        current = self.head
        while current is not None:
            count = count + 1
            total = total + current.budget
            print(f"{'count':<4}{current.name[:18]:<19}{current.phone:<12}"
                  f"{current.platform:<11}{current.service:<20}"
                  f"{current.budget:>8}")
            current = current.next
        print("-" * 74)
        print(f"Total Clients :", count, "      Total Monthly Budget : Rs.", total)

    # linear search by name or phone number, returns the node or None
    def find_client(self, key):
```

```

        current = self.head
    while current is not None:
        if current.name.lower() == key.lower() or current.phone == key:
            return current
        current = current.next
    return None

def search_client(self, key):
    node = self.find_client(key)
    if node is None:
        print("Sorry!", key, "is not available in the client list")
    else:
        print("Client Found!")
        print("Name      :", node.name)
        print("Phone     :", node.phone)
        print("Platform  :", node.platform)
        print("Service   :", node.service)
        print("Budget    : Rs.", node.budget)

def update_client(self, key, new_service, new_budget):
    node = self.find_client(key)
    if node is None:
        print("Sorry!", key, "is not available in the client list")
        return False
    node.service = new_service
    node.budget = new_budget
    print(node.name, "updated successfully!")
    return True

# remove a node by changing the link of the previous node
def delete_client(self, key):
    if self.head is None:
        print("--- No clients available ---")
        return False
    if self.head.name.lower() == key.lower() or self.head.phone == key:
        print(self.head.name, "deleted successfully!")
        self.head = self.head.next
        return True
    previous = self.head
    current = self.head.next
    while current is not None:
        if current.name.lower() == key.lower() or current.phone == key:
            previous.next = current.next
            print(current.name, "deleted successfully!")
            return True
        previous = current
        current = current.next
    print("Sorry!", key, "is not available in the client list")
    return False

# write every node to the file, one client per line
def save_to_file(self):
    with open(CLIENT_FILE, "w") as file:
        current = self.head
        while current is not None:
            file.write(f"{current.name},{current.phone},{current.platform},"
                      f"{current.service},{current.budget}\n")
            current = current.next

# read the file and build the linked list
def load_from_file(self):
    try:
        with open(CLIENT_FILE, "r") as file:
            for line in file:
                parts = line.strip().split(",")

```

```

        if len(parts) == 5:
            self.add_client(parts[0], parts[1], parts[2],
                            parts[3], int(parts[4]))
    except FileNotFoundError:
        print("(clients.txt not found - starting with an empty list)")

# check the username and password against login.txt
def check_login(username, password):
    try:
        with open(LOGIN_FILE, "r") as file:
            for line in file:
                parts = line.strip().split(",")
                if len(parts) == 2:
                    if parts[0] == username and parts[1] == password:
                        return True
    except FileNotFoundError:
        print("login.txt not found!")
    return False

def choose(title, options):
    for i in range(len(options)):
        print(f" {i + 1}. {options[i]}")
    while True:
        pick = input(title)
        if pick.isdigit() and 1 <= int(pick) <= len(options):
            return options[int(pick) - 1]
        print("Please enter a number from 1 to", len(options))

def ask_budget():
    while True:
        budget = input("Monthly Budget (Rs.) : ")
        if budget.isdigit() and int(budget) > 0:
            return int(budget)
        print("Budget must be a number greater than 0")

# ----- MAIN PROGRAM -----
print("=" * 50)
print("    SOCIAL MEDIA MARKETING AGENCY SYSTEM")
print("=" * 50)
user_name = input("Hello! Welcome --- What's your name : ")

logged_in = False
attempts = 3
while attempts > 0 and not logged_in:
    username = input(user_name + ", please enter Username : ")
    password = input(user_name + ", please enter Password : ")
    if check_login(username, password):
        logged_in = True
    else:
        attempts = attempts - 1
        print("<---- Incorrect Username or Password ---->")
        print("Attempts left :", attempts)

if not logged_in:
    print("Access denied. Please contact the agency admin.")
else:
    print("\n---- Login Successful ----\n")
    agency = AgencyList()
    agency.load_from_file()
    agency.display_clients()

```

```

choice = ""
while choice != "6":
    print("\n----- MENU -----")
    print(" 1. Add New Client")
    print(" 2. Display All Clients")
    print(" 3. Search Client")
    print(" 4. Update Client Package")
    print(" 5. Delete Client")
    print(" 6. Exit")
    choice = input("Enter your choice : ")

    if choice == "1":
        name = input("Client / Business Name : ").replace(", ", " ").strip()
        if name == "":
            print("Name cannot be empty")
        elif agency.find_client(name) is not None:
            print("Client", name, "already exists")
        else:
            phone = input("Phone Number (10 digits) : ")
            while not (phone.isdigit() and len(phone) == 10):
                phone = input("Invalid! Enter 10 digit Phone Number : ")
            print("Select Platform :")
            platform = choose("Platform number : ", PLATFORMS)
            print("Select Service :")
            service = choose("Service number : ", SERVICES)
            budget = ask_budget()
            agency.add_client(name, phone, platform, service, budget)
            agency.save_to_file()
            print("<---- Client stored successfully in clients.txt ---->")

    elif choice == "2":
        print("----- ALL CLIENTS -----")
        agency.display_clients()

    elif choice == "3":
        key = input("Enter Client Name or Phone Number to search : ")
        agency.search_client(key)

    elif choice == "4":
        key = input("Enter Client Name or Phone Number to update : ")
        if agency.find_client(key) is None:
            print("Sorry!", key, "is not available in the client list")
        else:
            print("Select New Service :")
            service = choose("Service number : ", SERVICES)
            budget = ask_budget()
            if agency.update_client(key, service, budget):
                agency.save_to_file()

    elif choice == "5":
        key = input("Enter Client Name or Phone Number to delete : ")
        if agency.delete_client(key):
            agency.save_to_file()

    elif choice == "6":
        print("----- THANK YOU FOR USING THE AGENCY SYSTEM -----")

    else:
        print("Please enter a valid option (1-6)")

```

OUTPUT

1) LOGIN

LOGIN SUCCESSFUL – CLIENTS LOADED FROM FILE

The first password is wrong, so the attempts left are shown. The second attempt matches a line in login.txt and the saved clients are loaded from clients.txt into the linked list.

```
=====
SOCIAL MEDIA MARKETING AGENCY SYSTEM
=====
Hello! Welcome --- What's your name : Riya
Riya, please enter Username : admin
Riya, please enter Password : admin12
<---- Incorrect Username or Password ---->
Attempts left : 2
Riya, please enter Username : admin
Riya, please enter Password : admin123

---- Login Successful ----
```

NO	NAME	PHONE	PLATFORM	SERVICE	BUDGET
1	TrendWave Apparel	9822334455	Instagram	Reels Editing	12000
2	Patil Sweets	9876543210	Facebook	Post Design	5000
3	FitZone Gym	9123456780	YouTube	Ad Campaign	18000

```
-----
Total Clients : 3      Total Monthly Budget : Rs. 35000
```

ACCESS DENIED AFTER 3 WRONG ATTEMPTS

The program stops after three wrong passwords.

```
=====
SOCIAL MEDIA MARKETING AGENCY SYSTEM
=====
Hello! Welcome --- What's your name : Riya
Riya, please enter Username : admin
Riya, please enter Password : admin1
<---- Incorrect Username or Password ---->
Attempts left : 2
Riya, please enter Username : admin
Riya, please enter Password : Admin123
<---- Incorrect Username or Password ---->
Attempts left : 1
Riya, please enter Username : admin
Riya, please enter Password : smm2025
<---- Incorrect Username or Password ---->
Attempts left : 0
Access denied. Please contact the agency admin.
```

2] ADDING A NEW CLIENT

An invalid phone number is asked again. Cafe Aroma is linked as the last node of the list and clients.txt is saved.

```
----- MENU -----
1. Add New Client
2. Display All Clients
3. Search Client
4. Update Client Package
5. Delete Client
6. Exit
Enter your choice : 1
Client / Business Name : Cafe Aroma
Phone Number (10 digits) : 98765
Invalid! Enter 10 digit Phone Number : 9000011111
Select Platform :
  1. Instagram
  2. Facebook
  3. YouTube
  4. LinkedIn
Platform number : 1
Select Service :
  1. Post Design
  2. Reels Editing
  3. Ad Campaign
  4. Account Management
Service number : 3
Monthly Budget (Rs.) : 8000
<---- Client stored successfully in clients.txt ---->
```

3] DISPLAYING ALL CLIENTS

The list is traversed from the head to the last node, and the total monthly budget is added up on the way.

```
----- MENU -----
1. Add New Client
2. Display All Clients
3. Search Client
4. Update Client Package
5. Delete Client
6. Exit
Enter your choice : 2
----- ALL CLIENTS -----
NO  NAME                PHONE        PLATFORM    SERVICE        BUDGET
-----
1   TrendWave Apparel  9822334455   Instagram   Reels Editing   12000
2   Patil Sweets       9876543210   Facebook    Post Design     5000
3   FitZone Gym        9123456780   YouTube     Ad Campaign     18000
4   Cafe Aroma         9000011111   Instagram   Ad Campaign     8000
-----
Total Clients : 4      Total Monthly Budget : Rs. 43000
```

4] SEARCHING A CLIENT

CLIENT FOUND

Linear search works in any letter case: "fitzone gym" finds FitZone Gym.

```
----- MENU -----  
1. Add New Client  
2. Display All Clients  
3. Search Client  
4. Update Client Package  
5. Delete Client  
6. Exit  
Enter your choice : 3  
Enter Client Name or Phone Number to search : fitzone gym  
Client Found!  
Name      : FitZone Gym  
Phone     : 9123456780  
Platform  : YouTube  
Service   : Ad Campaign  
Budget    : Rs. 18000
```

CLIENT NOT FOUND

No node has this phone number, so the search reaches the end of the list.

```
Enter your choice : 3  
Enter Client Name or Phone Number to search : 91230000000  
Sorry! 91230000000 is not available in the client list
```

5] UPDATING A CLIENT PACKAGE

Patil Sweets is found by phone number. Its service changes from Post Design to Account Management and its budget from Rs. 5000 to Rs. 15000.

```
----- MENU -----  
1. Add New Client  
2. Display All Clients  
3. Search Client  
4. Update Client Package  
5. Delete Client  
6. Exit  
Enter your choice : 4  
Enter Client Name or Phone Number to update : 9876543210  
Select New Service :  
1. Post Design  
2. Reels Editing  
3. Ad Campaign  
4. Account Management  
Service number : 4  
Monthly Budget (Rs.) : 15000  
Patil Sweets updated successfully!
```

6] DELETING A CLIENT

TrendWave Apparel is the head node, so the head now points to the next node (Patil Sweets).

```
----- MENU -----
1. Add New Client
2. Display All Clients
3. Search Client
4. Update Client Package
5. Delete Client
6. Exit
Enter your choice : 5
Enter Client Name or Phone Number to delete : TrendWave Apparel
TrendWave Apparel deleted successfully!
```

7] AFTER UPDATE & DELETE – DISPLAY ALL CLIENTS

Patil Sweets shows its new package and TrendWave Apparel is no longer in the list.

```
Enter your choice : 2
----- ALL CLIENTS -----
NO  NAME                PHONE        PLATFORM    SERVICE                BUDGET
-----
1   Patil Sweets        9876543210   Facebook    Account Management     15000
2   FitZone Gym         9123456780   YouTube     Ad Campaign            18000
3   Cafe Aroma          9000011111   Instagram   Ad Campaign            8000
-----
Total Clients : 3      Total Monthly Budget : Rs. 41000
```

8] INPUT VALIDATION & EXIT

A duplicate client name, a wrong menu choice, a wrong platform number, a non-numeric and a zero budget, and deleting a client that does not exist are all handled.

```
Enter your choice : 1
Client / Business Name : patil sweets
Client patil sweets already exists

Enter your choice : 9
Please enter a valid option (1-6)

Enter your choice : 1
Client / Business Name : Green Leaf Organics
Phone Number (10 digits) : 9555512345
Select Platform :
  1. Instagram
  2. Facebook
  3. YouTube
  4. LinkedIn
Platform number : 7
Please enter a number from 1 to 4
Platform number : 4
Select Service :
  1. Post Design
  2. Reels Editing
  3. Ad Campaign
  4. Account Management
Service number : 1
Monthly Budget (Rs.) : abc
Budget must be a number greater than 0
Monthly Budget (Rs.) : 0
Budget must be a number greater than 0
Monthly Budget (Rs.) : 3000
<---- Client stored successfully in clients.txt ---->

Enter your choice : 5
Enter Client Name or Phone Number to delete : Sunrise Hotel
Sorry! Sunrise Hotel is not available in the client list

Enter your choice : 6
----- THANK YOU FOR USING THE AGENCY SYSTEM -----
```

9] clients.txt FILE AFTER THE RUN

The file is saved after every change, so it always matches the linked list.

```
Patil Sweets,9876543210,Facebook,Account Management,15000
FitZone Gym,9123456780,YouTube,Ad Campaign,18000
Cafe Aroma,9000011111,Instagram,Ad Campaign,8000
Green Leaf Organics,9555512345,LinkedIn,Post Design,3000
```

CONCLUSION

The Social Media Marketing Agency System is a simple yet effective command-line tool for managing the clients of a social media marketing agency. It demonstrates the practical application of the Linked List data structure for storing client records dynamically, along with file handling for permanent storage and a login checked from a file. Fundamental programming concepts such as loops, conditionals and user input handling are used throughout. The project allows the agency to add, display, search, update and delete client records efficiently and to know its total monthly budget at any time. The SWOT analysis shows that social media marketing is a fast-growing field with many opportunities but also high competition, so keeping client records organised is key to an agency's success.